Mark Scheme

Q1.

Question	Scheme	Marks	AOs
(a)	$H_0: p = 0.1 \quad H_1: p \neq 0.1$	B1	2.5
		(1)	
(b)	Use of $X \sim B(50, 0.1)$ implied by sight of one of awrt 0.0052 or awrt 0.9755 or awrt 0.0245	M1	3.4
	Critical regions $X = 0$ or $X \geqslant 10$	A1	1.1b
	$X = 0$ and $X \geqslant 10$ plus $P(X = 0) =$ awrt 0.0052 and $P(X \geqslant 10) =$ awrt 0.0245	A1	1.1b
	SC: Both CR correct with no probabilities and no distribution seen scores M0A1A0		
		(3)	
(c)	0.0297	B1ft	1.1b
		(1)	
(d)	15 is <u>in the critical region</u> therefore there is evidence to support the <u>**manager**</u>'s belief	B1ft	2.2b
		(1)	
			(6 marks)

Notes		
(a)	B1	For both hypotheses in terms of p or π. Connected to H_0 and H_1 correctly Condone 10% but not 10
(b)	M1	Using correct distribution to find the probability associated with one tail of the CR If the correct distribution is <u>stated</u> (may be seen in part(a)) allow for one tail of the correct CR or one of (awrt 0.025 or awrt 0.005 or awrt 0.975) seen connected to a correct probability statement
	A1	Lower CR $X = 0$ / $X < 1$ / $X \leqslant 0$/ [condone eg $P(X = 0)$ labelled as CR] Or Upper CR $X \geqslant 10$ or $X > 9$ [condone $P(X \geqslant 10)$ oe labelled as CR]
	A1	Both CR's correct with the relevant probabilities Allow $\cup$ for "and" and $X > 9$, $X < 1$, $X \leqslant 0$ **[do not allow** $P(X = 0)$ or $P(X \geqslant 10)$ oe**]** Allow CR in different form eg $(9, \infty)$, $[10, \infty)$
(c)	B1ft	awrt 0.0297 or 2.97% or ft for the sum of the probabilities in (b) for "their 2 critical regions" if seen. If none seen it must be awrt 0.0297 **SC M0 in (b) for a one tail test** Allow B1ft for their one tail CR in (b) eg 0.0338 or 0.0245 or 0.0579
(d)	B1ft	A correct statement about 15 and "their CR" or sight $P(X \geqslant 15) = 0.0000738...$ and comparison with "their 0.0245" **and** a compatible correct statement in context. eg There is evidence that there has been a change in the **<u>proportion/probability</u>** arriving **<u>late</u>** Condone increase rather than change Do not allow contradicting statements. NB No CR given in (b) then B0

Q2.

	Scheme	Marks	AO
(a)	Negative	B1	1.2
		(1)	
(b)(i)	Rainfall mm or Pressure	B1	2.2b
(ii)	hPa or Pascals or hectopascals or mb or millibars	B1ft	1.1b
		(2)	
(c)	$H_0 : \rho = 0 \quad H_1 : \rho \neq 0$	B1	2.5
	Critical value: $-0.361(0)$	M1	1.1b
	$r < -0.3610$ so significant result and there is evidence of a correlation between Daily Total <u>Sunshine</u> and Daily Maximum Relative <u>Humidity</u>	A1	2.2b
		(3)	
(d)	Humidity is high and there is evidence of correlation and $r < 0$ So expect amount of sunshine to be <u>lower</u> than the <u>average</u> for Heathrow(oe)	B1	2.2b
		(1)	
		(7 marks)	

	Notes
(a)	B1 for stating negative. "Negative skew" is B0 though
(b)(i)	B1 for mentioning "rainfall" (allow "rain" <u>or</u> "precipitation") <u>or</u> "pressure" (if more than 1 answer both must be correct) NB the other quantitative variable for Perth is: Daily Mean Wind Speed and scores B0 [Not allowed "wind speed" since $r = +0.15$ and in winter might expect wind to raise temp]
(ii)	B1ft for giving the correct units. If Daily Mean Wind Speed (kn) or knots "Wind speed" and "knots" would score B0B1 but any other variable scores B0B0
(c)	B1 for both hypotheses correct in terms of ρ M1 for the correct critical value compatible with their H_1: allow $\pm$ 0.361(0) If the hypotheses are 1-tail then allow cv of $\pm$ 0.3061 e.g. Alternative hypothesis with $r < \pm 0.377$ implies a one-tail test <u>or</u> H_0 and H_1 in words saying "H_0: there is no correlation, H_1: there is correlation" is two-tail If there are no hypotheses (or they are nonsensical) assume 2-tail so M1 for $\pm$ 0.361(0) A1 for a correct conclusion in context based on comparing -0.377 with their cv. Condone incorrect inequality e.g. $-0.3610 < -0.377$ as long as they reject H_0 Do not accept contradictory statements such as "accept H_0 so there is evidence of …" Can say "support for Stav's <u>belief</u>"(o.e.e.g. "claim") or "evidence of a correlation between <u>sunshine</u> and <u>humidity</u>" condone "negative correlation" or comments such as "if humidity is high amount of sunshine will be low"
(d)	B1 for stating <u>low</u> amount of sunshine (o. e.) **and** some reference to $r < 0$ or fog Check for the following 2 features: (i) **low** sunshine: allow $\leqslant$ 5 hrs (LDS mean for 2015 is 5.3, humidity 97% is 4.1, $\geqslant$97% is 3.1) (ii) **negative** correlation may be described in words e.g. "high humidity gives low sunshine" <u>or</u> **fog** (LDS says >95% humidity is foggy) so less sunshine

Q3.

Part	Working or answer an examiner might expect to see	Mark	Notes
(a)	$H_0 : \rho = 0$ $H_1 : \rho > 0$	B1	This mark is given for both hypotheses in terms of ρ found correctly
	For sample size 24 at the 5% level of significance, the critical value = 0.3438	M1	This mark is given for selecting a suitable critical value compatible with H_1
	$0.446 > 0.3438$, so reject H_0 There is evidence that the product moment correlation coefficient (pmcc) is greater than 0	A1	This mark is given for a correct conclusion stated
(b)	The value of the pmcc is close to 1 so there is a strong positive correlation	B1	This mark is given for a correct explanation about the strength of the correlation
(c)	$\log_{10} y = -1.82 + 0.89 \log_{10} x$	M1	This mark is given for a correct substitution of both c and m
	$y = 10^{-1.82 + 0.89 \log x}$	M1	This mark is given for dealing with logs to find an expression in terms of y
	$y = 10^{-1.82} \times 10^{0.89 \log x}$ $y = 10^{-1.82} \times 10^{(\log x)^{0.89}}$	M1	This mark is given for a method to find values for a and n
	$y = 0.015 \times x^{0.89}$	A1	This mark is given for find a correct value of $a = 0.015$
		A1	This mark is given for find a correct value of $n = 0.89$
			(Total 9 marks)

Q4.

Qu	Scheme	Marks	AO
(a)	$H_0 : \rho = 0 \qquad H_1 : \rho < 0$	B1	2.5
	Critical value: -0.6215 (Allow any cv in range $0.5 < \lvert cv \rvert < 0.75$)	M1	1.1a
	$r < -0.6215$ so significant result and there is evidence of a negative correlation between w and t	A1	2.2b
		(3)	
(b)	e.g. As temperature increases people spend more time on the beach and less time shopping (o.e.)	B1	2.4
		(1)	
(c)	Since r is close to -1 , it is consistent with the suggestion	B1	2.4
		(1)	
(d)	t will be the explanatory variable since sales are likely to depend on the temperature	B1	2.4
		(1)	
(e)	Every degree rise in temperature leads to a drop in weekly earnings of £171	B1	3.4
		(1)	
		(7 marks)	

	Notes
(a)	B1 for both hypotheses in terms of ρ M1 for the critical value: sight of ± 0.6215 or any cv such that $0.5 < \lvert cv \rvert < 0.75$ A1 must reject H_0 on basis of comparing -0.915 with -0.6215 (if $-0.915 < 0.6215$ is seen then A0 but may use $\lvert r \rvert$ o.e. which is fine) and mention "negative", "correlation/relationship" and at least "w" and "t"
(b)	B1 for a suitable reason to explain negative correlation using the context given. e.g. "As temperature drops people are more likely to go shopping (than to the beach)" e.g. "As temperature increases people will be outside rather than in shops" A mere description in context of negative correlation is B0 SO e.g. "As temperature increases people don't want to go shopping/buy clothes" is B0 e.g. "Less clothes needed as temp increases" is B0
(c)	B1 for a suitable reason e.g. "strong"/"significant"/"near perfect" "correlation", $\lvert r \rvert$ close to 1 and saying it is consistent with the suggestion. Allow "yes" followed by the reason.
(d)	B1 For identifying t and giving a suitable reason. Need idea that "w depends on t" or "w responds to t" or "t affects w" (o.e.) Allow t (temperature) affects the other variable etc Just saying " t is the independent variable" or "t explains change in w" is B0 N. B. Suggesting causation is B0 e.g. "t causes w to decrease"
(e)	B1 for a description that conveys the idea of rate per degree Celsius. Must have 171, condone missing "£" sign.

Q5.

Question	Scheme	Marks	AOs
(a)	e.g. It requires extrapolation so will be unreliable (o.e.)	B1	1.2
		(1)	
(b)	e.g. Linear association between w and t	B1	1.2
		(1)	
(c)	H_0: $\rho = 0$ H_1: $\rho > 0$	B1	2.5
	Critical value 0.5822	M1	1.1a
	Reject H_0		
	There is evidence that the product moment correlation coefficient is greater than 0	A1	2.2b
		(3)	
(d)	Higher $\bar{t}$ suggests overseas and not Perth…lower wind speed so perhaps not close to the sea so suggest **Beijing**	B1	2.4
		(1)	
			(6 marks)

Notes:

(a)
B1: for a correct statement (unreliable) with a suitable reason

(b)
B1: for a correct statement

(c)
B1: for both hypotheses in terms of ρ
M1: for selecting a suitable 5% critical value compatible with their H_1
A1: for a correct conclusion stated

(d)
B1: for suggesting Beijing with some supporting reason based on t or w
Allow Jacksonville with a reason based just on higher $\bar{t}$